

Task Manager

Project Report

Internship: App Development Internship

Task: Task 2

Application Name: Task Manager

Developer: Nora Nagiy Sayed Hafez

Technology: Flutter

Storage: Hive

State Management: BLoC / Cubit

1. Project Overview

Task Manager is a cross-platform mobile application developed using Flutter.

The application helps users organize and manage their tasks through a simple and responsive interface. Users can create, view, update, delete, search, filter, and complete tasks.

Task data is stored locally on the device using Hive, allowing the application to maintain data without requiring an internet connection.

2. Objectives

The main objectives of the project were to:

- Build a functional cross-platform Flutter application.
 - Implement complete CRUD operations.
 - Store application data locally.
 - Implement state management using Cubit.
 - Create a responsive user interface.
 - Provide search and filtering functionality.
 - Support Light and Dark Mode.
 - Preserve the user's theme preference after restarting the application.
-

3. Technologies Used

Flutter

Used to build the cross-platform mobile application and user interface.

Dart

Used as the main programming language.

Hive

Used as the local NoSQL database for storing tasks.

SharedPreferences

Used to save and restore the user's selected theme.

BLoC / Cubit

Used to manage application state and handle task operations.

Material 3

Used to build the application's user interface and components.

4. Main Features

The application provides:

- Create new tasks.
 - View saved tasks.
 - Edit existing tasks.
 - Delete tasks with confirmation.
 - Mark tasks as completed.
 - Set task due dates.
 - Search by task title or description.
 - Filter tasks by All, Pending, and Completed.
 - Light Mode.
 - Dark Mode.
 - Persistent theme preference.
 - Responsive user interface.
 - Local data persistence.
-

5. CRUD Operations

Create

Users can create a new task by entering a title, description, and due date.

The task is then stored in the local Hive database.

Read

Saved tasks are retrieved from Hive and displayed on the Home Screen.

Update

Users can edit existing task information and change the completion status.

Delete

Users can delete an existing task. A confirmation dialog is displayed before deletion.

6. Local Data Storage

Hive is used as the local database.

Each task contains:

- ID
- Title
- Description
- Creation Date
- Due Date
- Completion Status

The data remains available after closing and reopening the application.

7. State Management

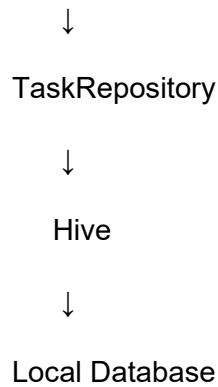
Cubit is used to manage the application's task state.

The application follows this general flow:

User Interface



TaskCubit



When a task is added, updated, completed, or deleted, the Cubit updates the application state and the UI reflects the changes.

8. Search and Filtering

The application includes a search field that allows users to search for tasks by:

- Task title
- Task description

Users can also filter tasks using:

- All
- Pending
- Completed

The number of tasks in each category is displayed next to the filter.

9. Theme Management

The application supports both Light Mode and Dark Mode.

Users can switch between the two modes using the button in the AppBar.

SharedPreferences is used to save the selected theme.

When the application is reopened, the previously selected theme is automatically restored.

10. Responsive Design

The application uses Flutter's responsive layout capabilities to adapt the interface to different screen sizes.

Spacing and layout components adjust according to the available screen width, providing a suitable experience on mobile devices and larger screens.

11. Application Screens

The project includes:

- Home Screen
 - Add Task Screen
 - Edit Task
 - Search
 - Task Filters
 - Delete Confirmation
 - Light Mode
 - Dark Mode
-

12. Testing

Test Case	Result
Create Task	Passd
View Tasks	Passd
Edit Task	Passd
Delete Task	Passd
Complete Task	Passd

Search Task	Passd
-------------	-------

Filter Tasks	Passd
--------------	-------

Set Due Date	Passd
--------------	-------

Light Mode	Passd
------------	-------

Dark Mode	Passd
-----------	-------

Save Theme Preference	Passd
--------------------------	-------

Local Data Persistence	Passd
------------------------	-------

Responsive Layout	Passd
-------------------	-------

13. Project Repository

The complete source code is publicly available on GitHub:

GitHub Repository:

<https://github.com/nouranagiy/task-manager-flutter>

The repository contains the Flutter source code, project structure, README documentation, and application screenshots.

14. Screenshots

The following screenshots demonstrate the main features of the application:

- Home Screen - Light Mode

- Home Screen - Dark Mode
 - Add Task
 - Edit Task
 - Search
 - Pending Tasks
 - Completed Tasks
 - Delete Confirmation
-

15. Challenges

During development, I worked on several practical challenges, including:

- Connecting the UI with local Hive storage.
 - Implementing CRUD operations.
 - Managing application state using Cubit.
 - Updating the UI after database changes.
 - Maintaining task data after application restart.
 - Implementing persistent Light/Dark Mode.
 - Creating a responsive layout for different screen sizes.
-

16. Conclusion

The Task Manager application provides a complete task management experience with local data persistence, CRUD operations, state management, search, filtering, responsive design, and persistent Light/Dark Mode.

The project provided practical experience in Flutter application development, local database integration, state management, and responsive user interface design.

17. Future Improvements

Possible future improvements include:

- Cloud synchronization.
- User authentication.
- Push notifications for due dates.
- Categories and task priorities.
- Cloud backup.
- Multi-device synchronization.